Memory and State Control for LLM-Driven NPC Dialogue: Scoped Dual-Tier Memory and Pre-Commit Invariant Validation under a Paired, Oracle-Audited Evaluation

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***Abstract*—Large language models let non-player characters hold conversations, but they forget early facts, propose changes that break game rules, and cost more as context grows. We present OBSIDIAN FLASK, a dialogue engine that separates generation from state control: a dual-tier memory whose retrieval is scoped to the current location and character, and an event-sourced world state in which every proposed change must pass a symbolic validation barrier before commit. We compare it with six memory baselines at an equal budget, including the retrieval rule of the base paper, Generative Agents, in a memory-by-control factorial with eight ablations, on seven worlds and four models, and audit every committed change with an invariant oracle that shares no code with the validator. With a 7-billion-parameter backbone, it recalled 63.7% of facts 20 or more turns old on held-out worlds, against 0.0% for a rolling context and 54.2% for the base paper's rule, but less than flat vector memory (69.3%) and no more than its earlier version (63.3%). The barrier committed no violation of the eight audited invariants in probes or live play and accepted none of 26,273 unlawful random proposals, against 4.4% attack success under direct writes, but it cannot stop secrets leaked in dialogue, and it rejected 20.0% of lawful requests that named entities by display name. Recovery was exact after 300 crashes simulated at random commit points. These guarantees are empirical and limited to the tested invariants, worlds and models.**

***Index Terms*—Large language models, non-player characters, long-term memory, event sourcing, runtime verification.**

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Source code, benchmark and run outputs: https://github.com/parthmital/RAG-Driven-NPC-Narrative-Engine.